Incident Command System — Full Project Plan

Nov 9, 2026 · @Rashid

This plan turns the working hi-fi prototype into an internal, air-gapped operational system for Ajman Civil Defence: Figma design first, then a web platform served from an internal server to PCs, iPads and phones, with on-premises AI and a live link to the iCCC operations-room software.

1. Summary

The system gives everyone running an incident one live picture: the operations room records the call and dispatches stations; the incident commander sectorises the site and sets the tactical mode; sector commanders report status and request support; the assistant commander approves and allocates. Every action is time-stamped automatically. Terminology follows the department SOP (numbered sectors, support sectors, tactical benchmarks, safety-priority messages).

What exists today. A single-page hi-fi prototype, tested on five device views (ops PC, IC iPad, AIC iPad, sector phone, sector iPad), synchronised through a cloud database, with AI for voice reports and photo analysis. The exercise rosters for Al Jurf and Administration stations are loaded.

What this plan delivers. The same product rebuilt as an internal platform: designed in Figma, served from a server inside the Civil Defence network with no internet dependency, with on-premises AI models and a two-way link to the iCCC operations-room software.

The one gating decision. The server platform and network team must be confirmed before Phase 3 (where the server lives, who administers it, which MDM manages the iPads, and the iCCC vendor contact for the integration). Everything after Phase 2 depends on it.

Item	Figure
Phases	7 (Figma → Pilot)
Duration to pilot in one station + ops room	about 9 months
Duration to full rollout	about 16 months
Team	8–9 people (about 100 person-months)

Item	Figure
Internet dependency in production	none

2. Phase 1 — Figma design

The prototype is the brief: every screen already exists and has been validated with the stations, so Figma formalises it rather than inventing it. Goal: a design system and pixel-accurate screens the developers build from without guessing.

Figma file structure (one file, one page per item):

1. **Foundations** — colour tokens (light and dark), Arabic type scale (Readex Pro for display, IBM Plex Sans Arabic for body), spacing, radii, elevation, the status colours (SOP benchmarks: in progress, all clear, under control, loss stopped, safety priority).
2. **Components** — buttons (large touch targets for gloves), pills, steppers, request cards, sector cards, KPI tiles, the evacuation banner, the SOP vests, the site-plan legend, map markers. Each as a Figma component with variants (state, size, severity) and Auto Layout.
3. **Operations Room — PC 1920×1080** — incidents list, dispatch map with real tiles, incident detail with status change and severity, command-team appointment, field station requests, stations board.
4. **Incident Commander — iPad 1180×820 landscape** — KPIs, tactical mode and evacuation, live site plan, support requests, agencies, media with AI analysis, log.
5. **Assistant IC — iPad landscape** — six views: Resources & requests, Sectors & teams (including support sectors), Force & casualties, External agencies, Notes, Local plan.
6. **Sector Commander — Phone 390×844 and iPad** — sector card with tactical mode controls, support request, voice report, camera, crew and drivers, command team.
7. **Overlays and states** — evacuation alarm, offline banner, safety-priority alert, empty and error states, RTL edge cases.
8. **Prototype flows** — clickable: call → dispatch → sectorise → request → approve → confirm receipt → close.

Rules for the handoff

- RTL is the base direction of every frame; mirrored icons are marked.
- Every text layer uses a token; no hard-coded colours or sizes.
- Component names match the code component names agreed in Phase 3 (for example `RequestCard`, `SectorPill`, `KpiTile`) so Figma Dev Mode maps one to one.
- Dark mode is a variable mode, not a second set of frames.

- Deliverables: Figma file, exported design tokens (JSON), icon set (SVG), an interactive prototype for the stations to review.

Exit gate: two sector commanders, one ops operator and one incident commander walk through the prototype and sign off the flows.

Week	Work
1	Foundations and tokens; audit of the prototype screens
2–3	Components and the Ops Room and IC screens
4–5	AIC views, Sector Commander phone and iPad, overlays
6	Prototype flows, review with the stations, sign-off

3. Functional specification

Everything below exists in the prototype and is the baseline scope for the production system. Each row is a feature; the rule column records the behaviour the stations agreed.

3.1 Operations Room (PC)

Feature	Behaviour and rules
New call	Fields: name, type (warehouse fire, hazmat fire, RTC, rescue…), coordinates, nearest landmark, description, trapped, injured, hazmat, critical notes, first station. Creates incident number `YYYY-NNNN`.
Severity	Four levels: minor, moderate, serious, catastrophe. Shown as a pill on every screen.
Incident status (iCCC model)	Ten statuses with change log: received, dispatched, en route, on scene, under control, leaving scene, returned, closed, cancelled, false alarm. Each change is time-stamped.
Station dispatch	Assign stations; each assignment creates a sector (numbered clockwise from the main entrance) staffed from the station roster. Up to 4 sectors.

Feature	Behaviour and rules
Dispatch map	Real map tiles (streets, satellite, dark). Station markers with distance and ETA to the incident; the incident marker; a station can be dispatched from the map.
Field station requests	Requests for extra stations from the IC arrive here; the operator picks a station and sends it, or rejects.
Command team	Appoints the IC and AIC; sees the sector commanders, deputies, safety officers, entry-control officers and OIC appliances.
Tactical mode	Reads the IC's mode (offensive/defensive) and sees the evacuation order. Can issue an evacuation.
Stations board	Each station: available or on which incident.
Close incident	Closes with the time; the log is kept.

3.2 Incident Commander (iPad)

Feature	Behaviour and rules
KPIs	Minutes since call, personnel on scene and sectors, personnel inside the building, pending requests, safety-priority messages, highest risk, injured, fatalities.
Tactical mode	Offensive or defensive for the whole incident; changing it resets every sector to the incident mode and is logged.
Per-sector mode (SOP 2.3.3)	A sector commander may switch their sector to defensive at once (IC is informed). A request to go offensive while the incident is defensive waits for the IC's approval here.
Evacuation	One button; every connected screen shows a full-screen alarm with sound until that device acknowledges. Cancel from the same place.
Live site plan	Site scene with sectors, fire location, assembly point, appliances on the main road and crew dots; tap a sector for its details.

Feature	Behaviour and rules
Support requests	All sector requests with status (pending, approved/en route, arrived, confirmed, rejected).
Extra station	Request one or more extra stations from the ops room with a reason.
External agencies	Police, ambulance, utilities… with representative and time added by the AIC.
Media	Photos and videos from sectors with the sector commander's voice note; AI analysis summary, hazards, suggested actions and SOP-referenced recommendation. The decision stays with the IC.
Event log	Every action, time-stamped, newest first.

3.3 Assistant Incident Commander (iPad)

View	Behaviour and rules
Resources & requests	Approve or reject sector requests (urgent first); allocate from the resource pool to sectors by stepper or drag; adjust the pool; a request becomes "arrived — awaiting confirmation" until the sector confirms receipt.
Sectors & teams	Number of sectors (2–4); station, sector commander and deputy (from an officers list); teams in the sector (type, leader, size); sector status by SOP benchmark; risk 1–5; support sectors (water supply, BA, decontamination) with their commanders; site plan selection.
Force & casualties	Personnel per sector, inside/outside, by task (entry, water supply, hoses, in teams), per station; appliances per type and sector; casualties: civilians injured, fatalities, firefighter injuries, trapped, rescued.
External agencies	Add agency, representative, notes.
Notes	Time-stamped journal visible to the IC and ops room.

View	Behaviour and rules
Local major-fire plan	Declare gold, silver and bronze commanders; a banner appears on every screen.

3.4 Sector Commander (phone and iPad)

Feature	Behaviour and rules
Sign in as sector	Choose the sector; the device then follows that sector only.
Sector card	Sector number, side of the building, station, commander, deputy, safety officer, radio call sign ("Sector N Commander" to "Incident Command"), status, risk, tactical mode.
Tactical mode controls	"Switch sector to defensive now" or "Request offensive mode" (waits for IC approval).
Support request	One tap per resource type; normal or urgent; status tracking; "Confirm receipt" when it arrives.
Voice report	Speak in dialect; speech-to-text; AI turns it into structured requests, a status update and a log line; the commander reviews before sending.
Photo and video	Camera capture, optional voice note converted to text, sent to the IC; AI analysis returned.
Crew	Every firefighter with task and position (entrance, inner corridor, upper floors, stairs, roof, outer face, supply point, staging); drivers with their appliances.
Command team	Can appoint every role except the IC and AIC.

3.5 Cross-cutting

Feature	Behaviour and rules
Real-time sync	Every change appears on all devices within seconds.

Feature	Behaviour and rules
Offline	A device keeps working without network and sends its queued changes when it reconnects.
Device presence	Each screen shows which roles are online.
Station rosters	Per-station default staffing (11 firefighters, 1 engine, 1 tanker, 1 supply vehicle) and the exercise rosters: leaders, drivers, firefighter groups, rescue teams; all editable.
Simulation mode	For training: generates events automatically. Disabled in production by default.
Multi-incident	Several incidents open at once; the ops room switches between them; field devices follow one.
Language and layout	Arabic RTL throughout; light and dark modes; one interface per device via a link parameter.

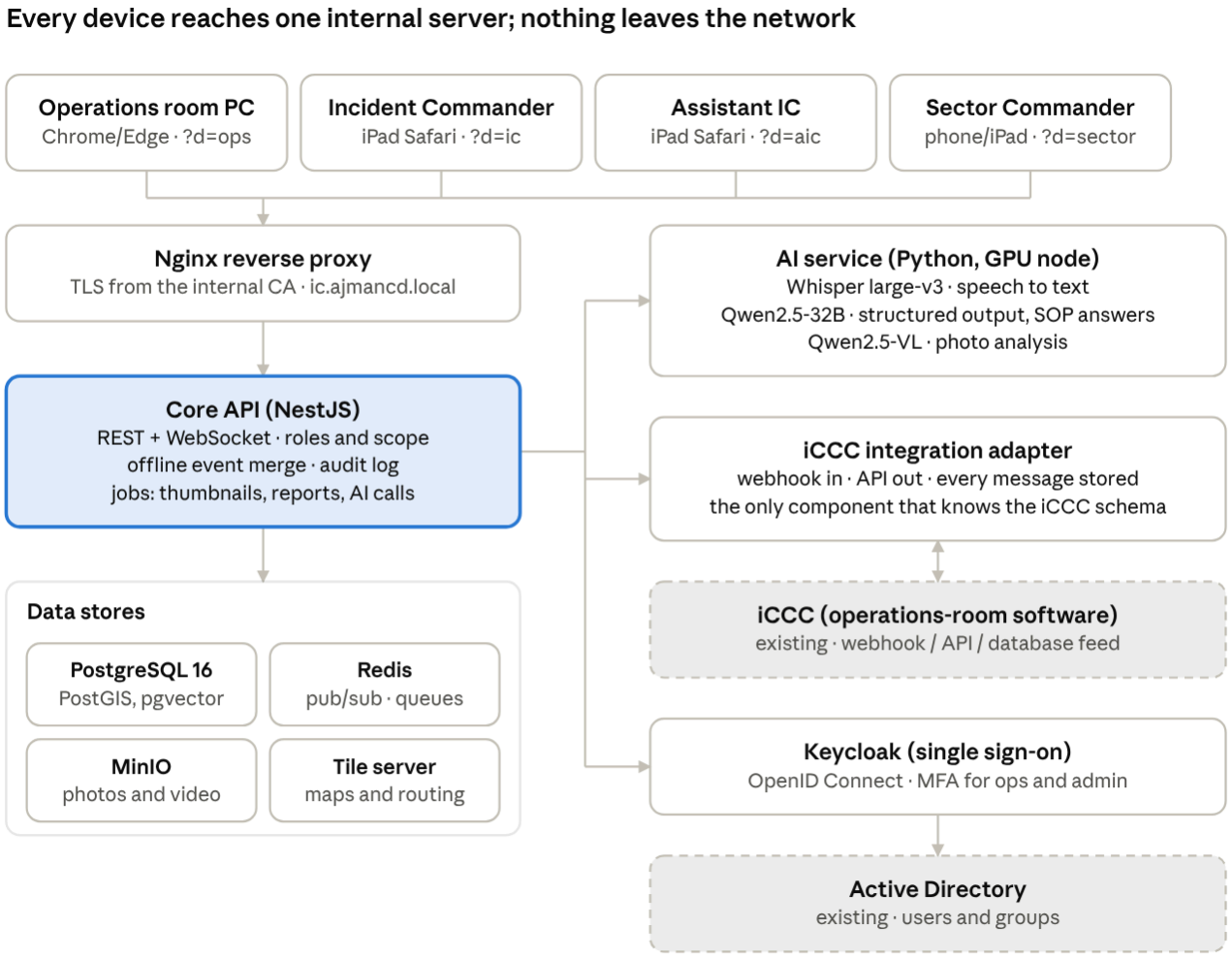
4. Technical specification

Area	Requirement	How it is met
Real-time sync	Any change reaches every connected device in under 2 s on the LAN	WebSocket channel per incident; the server pushes change events; clients apply them to local state
Offline first	A sector device must keep working with no network for the duration of an incident and reconcile later	Local store on the device (IndexedDB); every write is an append-only event with a device id and a monotonic clock; the server merges by event order, last-writer-wins per field, with the log kept in full
Identity	Every action is attributed to a person and a role	Single sign-on against the internal directory (Active Directory / LDAP); roles: operator, IC, AIC, sector commander, supervisor, admin

Area	Requirement	How it is met
Authorisation	Sector devices change only their sector; IC and AIC can change the incident; only the ops room creates and closes incidents	Role and scope checks on every API call and every WebSocket event
Audit log	Immutable, time-stamped, who did what	Append-only `events` table; no update or delete on it; nightly export
Media	Photos and video from phones, up to 20 MB, kept with the incident	Object storage on the server (MinIO, S3-compatible); thumbnails generated server-side
Maps	Real map tiles with no internet	Self-hosted tile server with UAE tiles pre-downloaded (OpenStreetMap data, optional satellite tiles licensed by the department); routing on the server for ETAs
Notifications	Evacuation and safety-priority alerts must reach every screen with sound	WebSocket event with high priority; the client shows the alarm and plays the sound; the alarm persists until acknowledged
Voice and camera	Browser access to microphone and camera on iPad and phone	Requires HTTPS (an internal certificate authority) and user permission; capture through the browser Media APIs
Speech-to-text	Emirati dialect, noisy environment	On-premises model (Section 9) running on the server; audio sent over the LAN, text returned
AI assistance	Voice to structured requests, photo analysis, SOP recommendation	On-premises models behind one internal AI service; the application never calls the internet
Reporting	After-incident timeline and response times	Generated from the events table; export to PDF
Performance	50 concurrent devices per incident, 10 incidents at once, with headroom	One application server and one database handle this; scale by adding a second application node behind the reverse proxy

Area	Requirement	How it is met
Availability	A failure must not stop an incident	Two application nodes, database with a streaming replica, nightly backups to a second machine; offline mode on the devices covers short outages
Browser support	Safari on iPad and iPhone, Chrome and Edge on Windows	Progressive Web App; tested on the actual department devices

5. Architecture



internal deployment · clients, server, AI node, existing systems

All devices talk only to the reverse proxy over HTTPS on the internal network. The core API holds the incident state and pushes changes over WebSocket; the AI service and the iCCC adapter are separate services the core calls, so either can be down without stopping an incident. Dashed boxes are systems that already exist in the department.

6. Technology choices

Frontend: TypeScript with React, built with Vite, as a Progressive Web App. The prototype is already browser-based and one codebase already serves five device layouts; React keeps that model and adds component reuse, typed data and a large hiring pool in the UAE. A PWA installs on the iPad and phone home screen from the internal server with no App Store and no Apple developer account, and gives offline caching and camera and microphone access through the browser.

- UI: React 18 + TypeScript, Tailwind CSS with the Figma tokens, Radix UI primitives for accessible controls.
- State and sync: TanStack Query for server data, a small event store on top of IndexedDB (Dexie) for offline writes.
- Maps: MapLibre GL JS against the self-hosted tile server; the site plan stays as SVG and Canvas as in the prototype.
- Rejected: React Native or Flutter (needs App Store or MDM app distribution and two codebases for web and mobile); Angular (heavier, smaller local talent pool for this scale).

Backend: Node.js with TypeScript (NestJS). One language across frontend and backend, first-class WebSocket support, and the prototype logic ports directly.

- API: REST for commands and queries (OpenAPI-documented), WebSocket (Socket.IO) for live events.
- Auth: OpenID Connect against Keycloak, which federates to Active Directory.
- Jobs: BullMQ on Redis for thumbnails, AI calls and report generation.
- Integration adapter: a separate small service that speaks to iCCC (Section 10), so the core never depends on the vendor's format.
- Rejected: .NET (good fit if the department's IT standard is Microsoft; take it instead of Node if so; everything else in this plan stays the same); Python/Django for the core (kept for the AI service only).

Database: PostgreSQL 16. Relational data (incidents, sectors, requests) with JSONB for flexible fields such as per-station rosters, `PostGIS` for coordinates and distances, logical replication for the standby, and a mature backup story. Redis for pub/sub and queues. MinIO for media.

- Rejected: MongoDB (no relational integrity for incident → sector → request); Firebase (cloud only, cannot run air-gapped).

Infrastructure. Everything in Docker containers orchestrated by Docker Compose on one host for the pilot and by two hosts (or a small Kubernetes such as k3s) for production. Nginx as reverse proxy and TLS termination with the internal CA. Prometheus and

Grafana for monitoring, Loki for logs.

Layer	Choice	Alternative if IT standard differs
Frontend	React + TypeScript PWA	Vue 3 + TypeScript
Backend	NestJS (Node.js)	ASP.NET Core
Database	PostgreSQL 16 + PostGIS	SQL Server (loses PostGIS; use its spatial types)
Cache and queue	Redis	RabbitMQ for queues
Media storage	MinIO	Windows file share behind the API
Identity	Keycloak → Active Directory	Direct AD FS / OIDC
Maps	MapLibre + self-hosted tiles	ArcGIS Enterprise if the department already licenses it
AI runtime	Python FastAPI service with GPU	Same, on CPU with smaller models

7. Database structure

PostgreSQL, one schema. Every table has `id` (UUID), `created_at`, `updated_at`; every change also lands in `events`, which is the audit log and the source for replays and reports. Text columns are UTF-8 Arabic; coordinates are PostGIS `geography(Point)`.

Table	Key columns	Notes
`users`	`id`, `directory_id`, `display_name`, `rank`, `station_id`, `active`	Mirrored from Active Directory on login
`roles` / `user_roles`	`user_id`, `role` (operator, ic, aic, sector, supervisor, admin), `scope`	Scope limits a sector commander to a sector
`stations`	`id`, `name`, `code`, `location`, `phone`, `supervisor`	The six Ajman stations plus ad-hoc ones

Table	Key columns	Notes
`station_rosters`	`station_id`, `role_key`, `person_name`, `group`, `vehicle`, `sort`	Default leaders, drivers, firefighter groups, rescue teams; copied into a sector on dispatch
`resource_types`	`code`, `name_ar`, `icon`	ff, engine, tanker, supply, ladder, foam, hazmat, csu, amb, rescue
`station_resources`	`station_id`, `resource_type`, `count`	Default appliances per station
`incidents`	`id`, `number`, `name`, `type`, `severity`, `status`, `location`, `landmark`, `description`, `trapped`, `injured`, `hazmat`, `notes`, `tactical_mode`, `evac_on`, `evac_by`, `opened_at`, `closed_at`, `iccc_ref`	`iccc_ref` links to the iCCC case id
`incident_status_history`	`incident_id`, `status`, `at`, `by`	The ten iCCC statuses
`incident_stations`	`incident_id`, `station_id`, `dispatched_at`, `arrived_at`, `released_at`, `oic_name`	
`incident_team`	`incident_id`, `role` (ic, aic, gold, silver, bronze), `person`, `appointed_at`, `by`	
`sectors`	`incident_id`, `number` (1–4), `side`, `station_id`, `commander`, `deputy`, `safety_officer`, `entry_control`, `status` (benchmark), `risk`, `tactical_mode`, `mode_request`, `target_object`	One row per sector
`sector_crew`	`sector_id`, `name`, `task`, `group`, `position`, `inside`	Position drives the "inside the building" count
`sector_drivers`	`sector_id`, `name`, `vehicle`	

Table	Key columns	Notes
`sector_teams`	`sector_id`, `type`, `leader`, `size`, `members`, `task`	Rescue and other teams
`sector_resources`	`sector_id`, `resource_type`, `count`	Allocated appliances
`incident_pool`	`incident_id`, `resource_type`, `count`	Unallocated resources
`support_sectors`	`incident_id`, `kind` (water, ba, decon), `commander`, `active`	SOP support sectors
`requests`	`incident_id`, `sector_id`, `resource_type`, `qty`, `priority`, `status` (new, approved, arrived, confirmed, rejected), `requested_at`, `decided_at`, `decided_by`, `eta`, `confirmed_at`	
`station_requests`	`incident_id`, `count`, `reason`, `status`, `requested_by`, `decided_by`, `station_id`	Extra-station asks to the ops room
`agencies`	`incident_id`, `name`, `representative`, `note`, `added_at`	
`casualties`	`incident_id`, `civ_injured`, `fatalities`, `ff_injured`, `trapped`, `rescued`, `updated_at`	One row per incident, history in `events`
`journal`	`incident_id`, `author`, `text`, `at`	AIC notes
`media`	`incident_id`, `sector_id`, `kind`, `object_key`, `thumb_key`, `note_text`, `duration`, `taken_at`, `by`	Files live in MinIO; `object_key` points to them
`media_analysis`	`media_id`, `summary`, `hazards`, `actions`, `mode_suggestion`, `sop_refs`, `model`, `at`	AI output, kept for audit

Table	Key columns	Notes
`voice_reports`	`incident_id`, `sector_id`, `audio_key`, `transcript`, `parsed`, `accepted`, `at`	
`events`	`id`, `incident_id`, `sector_id`, `actor_id`, `role`, `device_id`, `type`, `payload` (JSONB), `client_ts`, `server_ts`	Append only; the event log and the offline merge source
`devices`	`id`, `user_id`, `role`, `sector_id`, `last_seen`, `app_version`	Presence
`sop_chunks`	`id`, `title`, `page`, `text`, `embedding` (pgvector)	The SOP manual split for retrieval
`iccc_messages`	`id`, `direction`, `iccc_case_id`, `incident_id`, `payload`, `received_at`, `processed_at`, `error`	Every message exchanged with iCCC

Relationships. `incidents` 1→n `sectors`, `requests`, `media`, `events`, `incident_stations`; `sectors` 1→n `sector_crew`, `sector_drivers`, `sector_teams`, `sector_resources`; `stations` 1→n `station_rosters`, `station_resources`. `requests.sector_id` and `media.sector_id` are nullable for incident-level items.

Indexes. `events(incident_id, server_ts)`, `requests(incident_id, status)`, `incidents(status, opened_at desc)`, `sop_chunks` vector index (HNSW), `media(incident_id, taken_at desc)`.

Retention. Incidents and events are kept indefinitely; media is kept for a period the department sets (proposal: 2 years), then archived to cold storage.

8. Devices on the internal network

The application is a web app served over HTTPS from the internal server, so iPads, phones and PCs need only a browser and a route to that server. No App Store, no public internet.

How a device reaches the server

1. The server gets an internal DNS name, for example `ic.ajmancd.local`, pointing to the reverse proxy.

2. The department's internal certificate authority issues a TLS certificate for that name. HTTPS is mandatory: Safari and Chrome allow camera, microphone and offline storage only on secure origins.
3. Field devices join a dedicated Wi-Fi SSID (WPA3-Enterprise with device certificates) at stations and headquarters. On scene they use the department's private mobile network or a vehicle router (4G/5G private APN or mesh) that tunnels into the internal network. With no coverage, the device works offline and syncs when back in range.
4. On first use the user opens the URL in Safari or Chrome and taps "Add to Home Screen"; from then on it opens full-screen like a native app.

Managing the devices (MDM)

- Enrol all iPads and iPhones in an MDM (Jamf, Microsoft Intune or Apple Business Manager with a local MDM). The MDM pushes the Wi-Fi profile, the internal CA root certificate (so the browser trusts the server), the home-screen web clip pre-configured to the role URL (`?d=sector`), and optional single-app or kiosk mode.
- PCs in the operations room get the CA certificate through Group Policy and a pinned browser shortcut to `?d=ops`.
- Device certificates issued by the MDM can also authenticate the device itself, in addition to the user's single sign-on.

Per-device requirements

Device	Minimum	Notes
Ops room PC	Windows 10/11, Chrome or Edge, 1920×1080 or wider	Second screen for the dispatch map recommended
IC and AIC iPad	iPadOS 16+, Safari, 10.9″ or larger, cellular model	Rugged case; connected to the vehicle router
Sector commander phone	iOS 16+ or Android 12+ (Chrome), cellular	Glove-friendly case; the PWA uses large touch targets
Sector commander iPad	iPadOS 16+	Same build as the phone, two-column layout

What offline means in practice. The PWA caches its code and the last known incident state; writes queue locally; the device shows an "offline — N changes waiting" banner; on reconnect the queue is sent in order and the screen refreshes. Evacuation alarms issued while a device was offline are shown the moment it reconnects.

Push notifications when the app is closed. Web push on iOS requires the PWA to be installed to the home screen and works on an internal server through a self-hosted push relay only partially; the reliable path is to keep the app open in kiosk mode on duty devices. If true background alerts are required, Phase 6 adds a thin native wrapper (Capacitor) distributed through the MDM, reusing the same web code.

9. AI fully offline

The prototype uses a cloud model for four jobs. Each has a strong open-weight model that runs on a server inside the network with no internet. All of them sit behind one internal AI service (Python, FastAPI) that the application calls over the LAN; swapping a model later changes nothing in the application.

Job in the prototype	Local model	Why this one	Runtime
Speech-to-text, Emirati dialect, noisy scene	**Whisper large-v3** (OpenAI, open weights) served with faster-whisper; fine-tune on 20–40 hours of recorded department radio and drills for the dialect and vehicle names	Best open multilingual accuracy on Arabic; dialect fine-tuning is a known, documented process; runs in near real time on one GPU	CTranslate2, GPU
Voice report → structured requests, status, log line	**Qwen2.5-32B-Instruct** or **Llama 3.3 70B** (quantised) with JSON-schema constrained output	Strong Arabic understanding and reliable JSON; constrained decoding guarantees the schema the app expects	vLLM or llama.cpp, GPU
Field photo analysis (summary, hazards, actions)	**Qwen2.5-VL-32B** (vision-language)	Open vision-language model with good Arabic output and strong scene description; replaces the cloud image analysis one to one	vLLM, GPU

Job in the prototype	Local model	Why this one	Runtime
SOP recommendation (retrieve the right SOP sections, cite page numbers)	**Retrieval**: `bge-m3` embeddings in pgvector over the 26 SOP chunks (expandable to the full manual) + the same LLM to write the recommendation with citations	Retrieval keeps answers grounded in the department's own SOP; the model cannot invent procedures	pgvector + LLM
Fallback when the GPU is down	Whisper medium and Qwen2.5-7B on CPU	Slower but keeps the features alive	llama.cpp, CPU

Rules carried over from the prototype. The AI never acts on its own: a voice report is shown to the sector commander before sending; a photo analysis and SOP recommendation are suggestions to the IC. Every AI output is stored with the model name and version for audit (`media_analysis`, `voice_reports`). If the AI service is unavailable, every other feature keeps working.

Hardware sizing (one AI node)

Profile	GPU	RAM	Covers
Pilot	1 × NVIDIA L40S 48 GB or RTX 6000 Ada	128 GB	Whisper large-v3 + Qwen2.5-32B (4-bit) + Qwen2.5-VL-32B (4-bit), 5–10 concurrent requests
Production	2 × L40S or 1 × H100 80 GB	256 GB	Full-precision 32B models or Llama 3.3 70B, 20+ concurrent requests, headroom for fine-tuning
Minimum (CPU only)	none	64 GB	Whisper medium + 7B model; voice report 5–10 s, photo analysis 20–30 s

Getting the models in. Weights are downloaded once on an internet-connected machine, checksum-verified, scanned, and carried in on approved media. The AI node has no route to the internet. Model updates follow the same path, quarterly.

Dialect fine-tuning plan. Record 30 drill sessions with consent; transcribe 20–40 hours; fine-tune Whisper on the department's terms (vehicle names, station names, ranks, SOP phrases); measure word error rate before and after on a held-out set; target under 15%

WER on radio-quality audio.

10. iCCC integration (operations room)

Goal: a call received in iCCC appears in the Incident Command System without retyping, and status changes made in either system reach the other. The approach does not depend on knowing iCCC's internals yet: a separate **integration adapter** service sits between the two, so the core system only ever sees its own event format.

Step 1 — discovery with the iCCC vendor (2 weeks). Ask for: the integration options the installed version supports (outbound webhooks, a REST or SOAP API, CAD/AVL message feed, database views, or file drop), the event catalogue (new call, dispatch, unit status, close), authentication method, and the test environment. The answer decides which of the three patterns below is used; often it is a mix.

Pattern	When iCCC offers	How it works	Latency
A. Webhook push (preferred)	Outbound HTTP callbacks on events	iCCC POSTs a signed JSON message to `https://ic.ajmancd.local/integrations/iccc/webhook`; the adapter verifies the signature, maps it to an incident event, and the core creates or updates the incident	under 2 s
B. API polling	A read API but no callbacks	The adapter polls `GET /calls?since=` every 5–10 s, de-duplicates by case id, and emits events	5–10 s
C. Database or file feed	Neither, but a reporting database or export	A read-only database view or a watched folder; the adapter reads new rows or files	10–60 s

Outbound (our system → iCCC). The adapter exposes the same options in reverse: when the IC changes the tactical mode, orders evacuation, confirms a station on scene, or the ops room closes the incident, the adapter calls iCCC's API (or writes to its inbound queue) to update the case. If iCCC has no write interface, the adapter writes a status note the dispatcher sees, and the dispatcher updates iCCC manually; the gap is logged.

Mapping (first version)

iCCC	Incident Command System
Case / call id	`incidents.iccc_ref`
Call type and priority	`incidents.type`, `incidents.severity` (table of mappings maintained by the ops supervisor)
Address and coordinates	`incidents.location`, `landmark`
Dispatched units	`incident_stations` and, per station, a sector staffed from the roster
Unit status (en route, on scene, available)	`incident_status_history`, `incident_stations.arrived_at`
Case closed	`incidents.status = closed`

Adapter design

- Its own small service (NestJS or .NET, matching the iCCC side), its own credentials, reachable only from the iCCC server and the core API; it is the only component that knows iCCC's schema.
- Every message in both directions is stored in `iccc_messages` before processing, so nothing is lost and every exchange is auditable. Processing is idempotent: the same message twice has no second effect.
- Security: mutual TLS between iCCC and the adapter where the vendor supports it, otherwise an HMAC signature header and an IP allow-list on the reverse proxy; secrets in the server vault, rotated.
- Failure handling: if the core is down, the adapter queues; if iCCC is down, the ops operator can still create the call manually and link it later by case id. A dashboard tile shows "iCCC link: live / delayed / down".
- Testing: a replay tool feeds recorded iCCC messages into the adapter in the test environment; go-live only after a full drill with both systems.

Sequence for a new call (pattern A)

1. Caller → iCCC: call logged, units proposed.
2. iCCC → adapter: webhook `call.created` with case id, type, location.
3. Adapter → core API: `POST /incidents` with `iccc_ref`; the ops room screen shows the new incident within 2 s, pre-filled.
4. Operator confirms stations in either system; `unit.dispatched` events keep both in step.

5. IC and sectors work in the Incident Command System; `incident.updated` events flow back as case notes and status.
6. Ops room closes the incident; adapter → iCCC `case.close`; both records carry the same timeline.

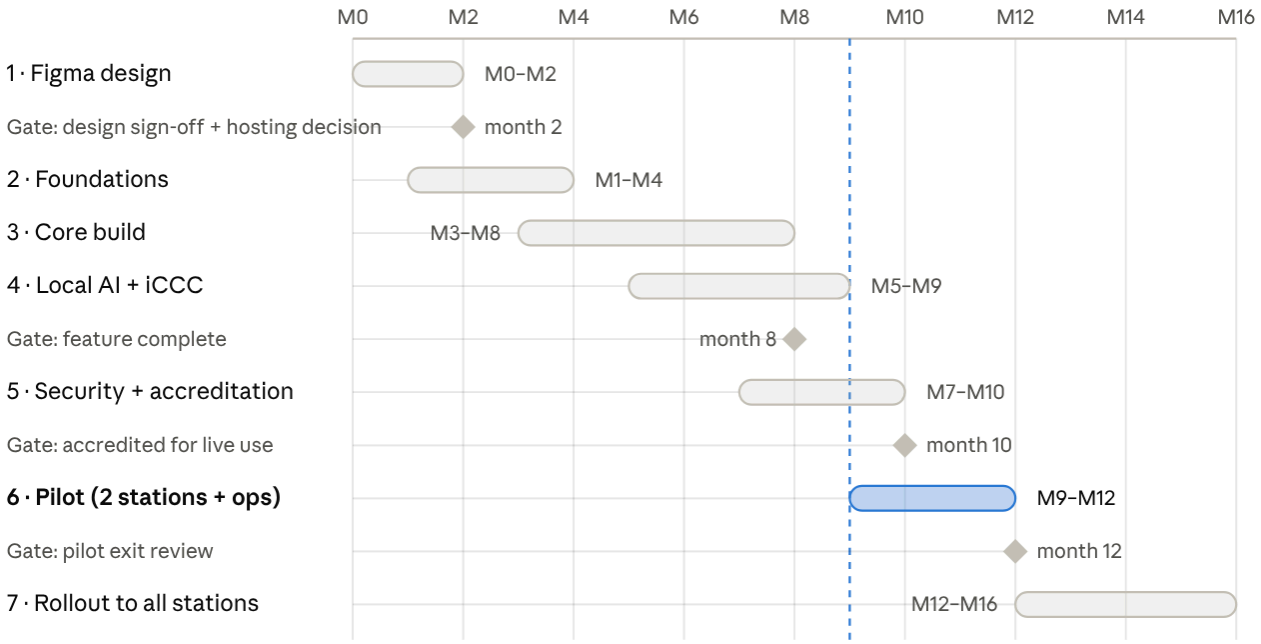
11. Security and compliance on an air-gapped network

- **No internet, by design.** The servers have no default route out. Map tiles, fonts, libraries and AI weights are bundled or hosted internally; the build pipeline runs on an internal CI runner with a vetted package mirror.
- **Identity.** Single sign-on against Active Directory through Keycloak; MFA for the ops room and admin roles if the department requires it; device certificates from the MDM for field devices.
- **Authorisation.** Role and scope on every call: a sector device can only change its sector; only operators create and close incidents; only the IC changes the incident's tactical mode; the adapter has a service account with the narrowest rights.
- **Transport.** TLS 1.2+ everywhere with the internal CA; HSTS; WebSocket over TLS.
- **Data at rest.** Disk encryption on the servers; PostgreSQL and MinIO on encrypted volumes; backups encrypted and stored on a second machine in a different room.
- **Audit.** Immutable `events` and `iccc_messages`; admin actions logged; monthly export to the department's records system.
- **Media privacy.** Photos and videos may show casualties: access limited to the incident's command team and supervisors; watermarking on export; retention policy agreed with legal.
- **Hardening.** Containers run as non-root with read-only file systems; monthly patching from the internal mirror; vulnerability scan of images before deployment; penetration test before the pilot.
- **Availability.** Two application nodes, database replica, documented restore drill once a quarter; offline mode covers device-side outages.
- **Compliance.** Align with the UAE Information Assurance Standards (NESA/TDRA IA) and the department's own IT policy; the security engineer owns the control mapping from Phase 3.

12. Delivery plan

Pilot starts in month 9, full rollout ends in month 16

Seven phases and four gates, in months from project start



delivery plan · 7 phases, 4 gates, 16 months

Phases overlap on purpose: foundations start while design finishes, and the AI and iCCC work runs beside the core build once the hosting and vendor decisions are in. The pilot runs at Al Jurf and Administration with the operations room for three months before rollout.

Phase	Months	Main deliverables	Exit gate
1 Figma design	1–2	Design system, all screens, clickable prototype	Stations sign off the flows; hosting, MDM and iCCC vendor decided
2 Foundations	2–4	Servers, internal CA, Keycloak + AD, CI runner, PostgreSQL, API skeleton, PWA shell, tile server	A device on the internal Wi-Fi signs in and sees an empty incident
3 Core build	4–8	Ops room, IC, AIC, sector views; real-time sync; offline queue; site plan; dispatch map; rosters; evacuation	Feature complete against Section 3; drill on the test server

Phase	Months	Main deliverables	Exit gate
4 Local AI + iCCC	5–9	AI service with the four local models, Whisper fine-tune, SOP retrieval; iCCC adapter with the vendor's pattern	Voice report WER under 15%; a test call flows from iCCC to the screen
5 Security + accreditation	7–10	Hardening, control mapping, penetration test, backup and restore drill, monitoring	Accreditation to operate on the internal network
6 Pilot	9–12	Two stations and the ops room on live incidents and drills; weekly fixes; training	Pilot exit review: response-time and adoption targets met
7 Rollout	12–16	Remaining stations, MDM enrolment of all devices, operations handbook, handover to IT	All stations live; support handed to the department

13. Team and effort

Role	Commitment	Months	Responsibility
Project manager	full	1–16	Plan, vendor and IT coordination, acceptance
Product designer (Figma)	full, then half	1–16	Design system, screens, usability tests
Frontend developer × 2	full	2–16	PWA, offline sync, site plan, maps
Backend developer	full	2–16	API, WebSocket, database, reports
Integration engineer	half	3–12	iCCC adapter, AD/Keycloak, MDM
AI engineer	half	4–12	Local models, Whisper fine-tuning, retrieval
Security and infrastructure engineer	half, full at accreditation	1–16	Servers, CA, hardening, backups, monitoring

Role	Commitment	Months	Responsibility
QA engineer	half	4–16	Test plans, device matrix, drill tests
Field reference officer	quarter	1–16	SOP correctness, drills, user training

About 100 person-months in total. Peak team: 8–9 people in months 4–10.

14. Risks and mitigations

Risk	Impact	Mitigation
Server platform and hosting decision slips	Delays every phase after design	Decide in month 1; pilot on one department server if the data centre is not ready
iCCC offers no webhook or API	Integration falls back to polling or file feed	Discovery in month 3; the adapter design supports all three patterns
Dialect speech recognition below target	Voice report unreliable	Fine-tune on department recordings; keep the typed fallback; measure WER each sprint
GPU procurement delay	AI features late	CPU fallback models for the pilot; order hardware in month 2
Field connectivity on scene	Devices offline for long periods	Offline-first design; vehicle routers; test in drills
Apple restrictions on PWAs (push, background)	Alerts missed when the app is closed	Kiosk mode on duty devices; Capacitor wrapper in Phase 6 if needed
Scope growth during pilot	Schedule and budget	Change board; only SOP-driven changes accepted in the pilot
Accreditation takes longer than planned	Go-live slips	Security engineer involved from month 1; control mapping done during build, not after

15. Open questions for the client

- [ ] Where will the servers live (department data centre, government cloud, or a server room at HQ), and who administers them?
- [ ] Which identity directory and MDM does the department run (Active Directory, Intune, Jamf)?
- [ ] iCCC vendor and version, and the contact for integration; does the contract include integration support?
- [ ] Is a Microsoft stack (.NET, SQL Server) mandated by IT policy? If yes, the alternatives column in Section 6 applies.
- [ ] Field connectivity on scene today: private mobile network, vehicle routers, or none?
- [ ] Can drill audio be recorded for dialect fine-tuning, and under which consent rules?
- [ ] Retention period for photos and videos, and who may view them after the incident.
- [ ] Which stations join the pilot after Al Jurf and Administration, and in what order?
- [ ] GPU budget: pilot profile (one L40S) or production profile from the start?